Hadi Mirza

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Education

University of California, Santa Cruz — Santa Cruz, CA
Bachelor of Science in Electrical Engineering — *Expected June 2027*

Relevant Coursework: Advanced Analog ICs, Analog Electronics + Lab, High-Speed Digital Design, Energy Generation & Control + Lab (Power Systems), Signals & Systems + Lab, Logic Design + Lab, Computer Systems & C Programming, Computer Systems & Assembly, Senior Capstone (in progress)

Technical Skills

Hardware Design: PCB Design (KiCad, Altium), Multi-layer/High-Speed Routing, Power Electronics, Analog Design
Simulation: LTspice, PSpice, PSCAD/EMTDC, PowerWorld, MATLAB/Simulink
Lab & Fab: Oscilloscopes, Function Generators, SMD/THT Soldering, Board Bring-up, JLCPCB, 3D Printing, Onshape
Languages & Platforms: C, C++, Python, Verilog, RISC-V Assembly, MATLAB; STM32, ESP32, Arduino, Xilinx Artix-7 (Vivado), Git, Linux

Experience

Electrical Team Member – Low Voltage & Power Electronics — Oct. 2024 – Present
Formula Slug (FSAE Electric), UC Santa Cruz — *Santa Cruz, CA*

- Leading design of the FS5 **Load Slammer**, a 50 A transient-load tester in KiCad used to characterize slew rate and transient response of the team's custom LV power supplies
- Re-architected the design from a hard-switched gate driver to a **closed-loop linear op-amp control** (OPA994 driving a 100 V / 1.7 mΩ MOSFET) for adjustable step size and slew rate
- Added a 4-terminal 5 mΩ Kelvin sense resistor, net-tied power/signal grounds, and 50 Ω-terminated coax I/O for clean oscilloscope current measurement
- Contributed to **BMS and shutdown-circuit** schematics and wiring in KiCad supporting safe emergency shutdown and FSAE rules compliance
- Developing an in-helmet driver audio board (MAX98357A I²S Class-D amp, condenser mic) for driver-to-pit radio

Projects

Ground Bounce EMI Detector | *Altium Designer, PCB Fabrication, Signal Integrity* — Summer 2026

- Designed a 2-layer Altium PCB to generate and measure ground bounce/EMI from switching logic, with SMA test points
- Hand-soldered all SMD components and captured ±20 mV ground ringing on a 5 V, 1 MHz edge using a 500 MHz Keysight oscilloscope

6-Layer FPGA Development Board | *High-Speed Routing, Impedance Control* — Summer 2026

- Routed a custom 6-layer FPGA board with dedicated power and ground planes to minimize crosstalk and EMI; I/O architecture modeled after the Basys 3
- Length-matched differential pairs and tuned trace impedances for 50 MHz signals; ran full DRC and schematic validation for manufacturability

POBO Silent Haptic Pillow Alarm | *KiCad, ESP32, Li-Ion Charging* — Fall 2026

- Designed an ESP32 carrier PCB driving 10 haptic motors via a logic-level MOSFET low-side switch with flyback protection, plus USB-C Li-ion charging (TP4056); iterated 5 revisions through JLCPCB

Whiteboard CNC Drawing Robot | *C/C++, Arduino, Stepper Control, PCB Design* — 2026 – Present

- Built a 2-axis X-Y plotter and wrote C/C++ firmware driving stepper motors through external drivers to draw precise geometric shapes
- Designing a custom PCB to consolidate power delivery and control; developing image-to-toolpath conversion

Real-Time VGA Video Game | *Verilog, Xilinx Artix-7, Vivado* — Fall 2025

- Implemented a hardware-only game on an Artix-7 FPGA with a VGA controller timed for 640×480 @ 60 Hz and hierarchical FSMs for game logic and pixel generation

Leadership

President, Multicultural Student Association, UC Santa Cruz — Lead 50+ members; organize events drawing 200+ attendees